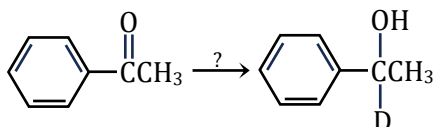




Reduction : RACE-01

Metal hydride reduction (LiAlH_4 ; NaBH_4 ; DIBAL-H; B_2H_6 ; LiBH_4 ; NaBH_3CN)

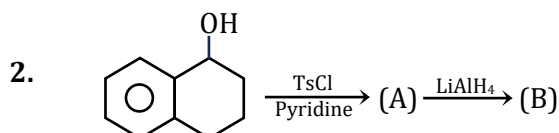
1. Which of the following reagents would carry out the following transformation ? ($\text{D} = 2\text{H}$)



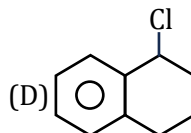
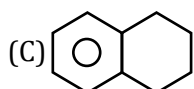
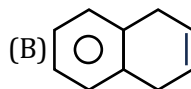
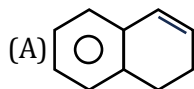
- (A) NaBD_4 in CH_3OH
 (C) NaBD_4 in CH_3OD

- (B) LiAlH_4 , then D_2O
 (D) LiAlD_4 , then D_2O

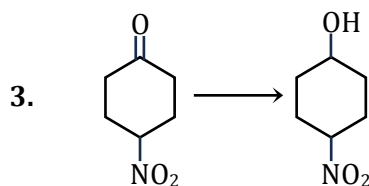
CRD120



Product (B) of the above reaction is :



CRD121

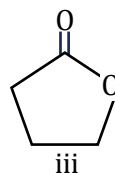
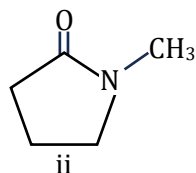
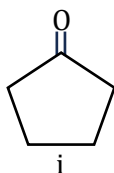


Above conversion can be achieved by :

- (A) LiAlH_4 (B) NaBH_4 (C) H_2/Ni (D) CrO_3

CRD122

4. Arrange the following compounds in order of their reactivity toward LiAlH_4 .



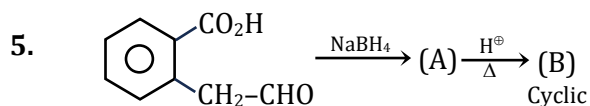
- (A) $\text{i} < \text{ii} < \text{iii}$

- (B) $\text{i} < \text{iii} < \text{ii}$

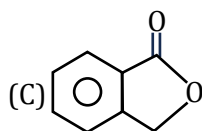
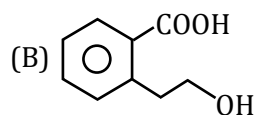
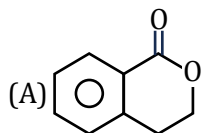
- (C) $\text{ii} < \text{i} < \text{iii}$

- (D) $\text{ii} < \text{iii} < \text{i}$

CRD123

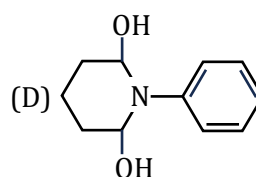
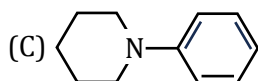
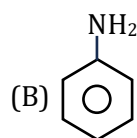
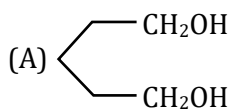
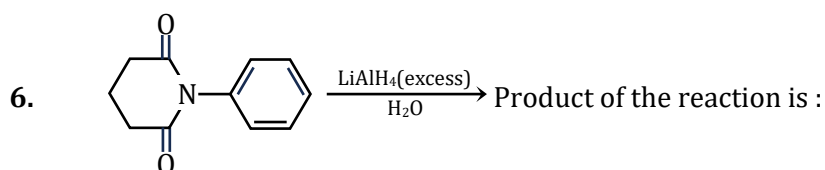


Compound (B) is :



(D) (A) and (C) both

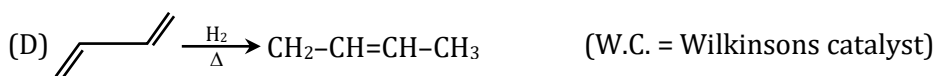
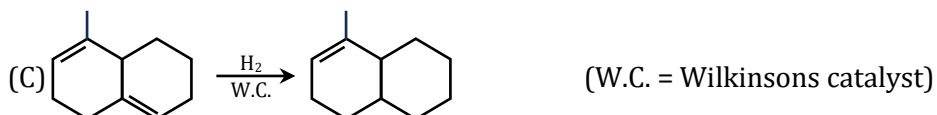
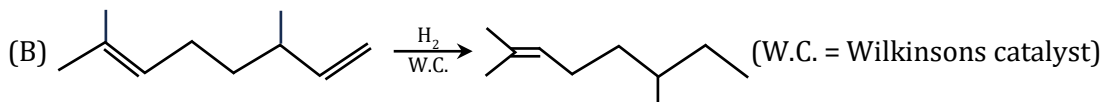
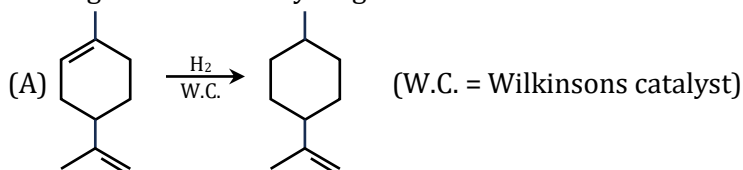
CRD124



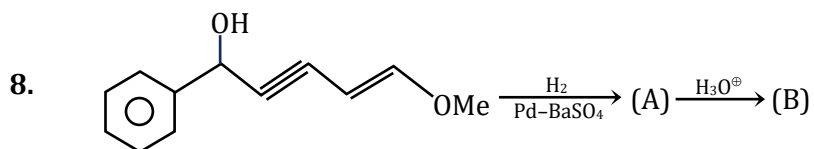
CRD128

Catalytic hydrogenation (Except alkene and alkyne)

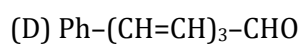
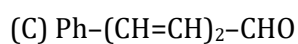
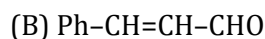
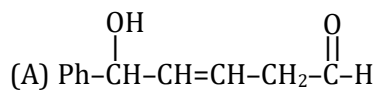
7. In the given selective hydrogenation which combination is incorrect ?



CRD118

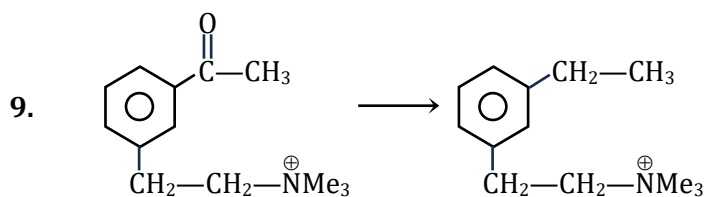


Product (B) is :

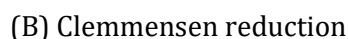
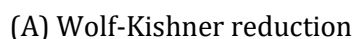


CRD119

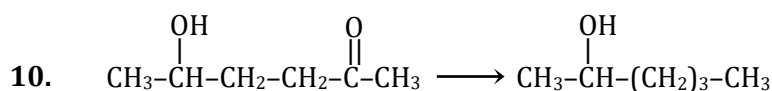
Bouvealt Blanc reduction



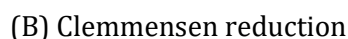
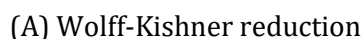
Above conversion can be achieved by :



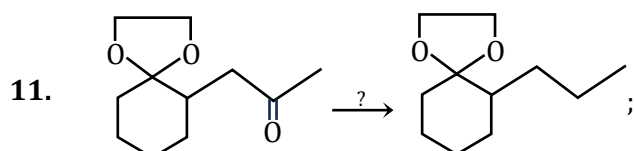
CRD125



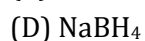
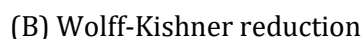
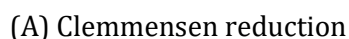
Above conversion can be achieved by :



CRD126



Above conversion can be carried out by :



CRD127

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	D	A	C	A	C	B	A
Question	11									
Answer	B									